Transparent Medical Image Classification via Explainable AI

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Abstract

Deep learning models for medical image classification often achieve high accuracy but lack interpretability, limiting their clinical adoption and trustworthiness. This study proposes a multi-method explainability framework combining Gradient-weighted Class Activation Mapping (Grad-CAM), SHapley Additive exPlanations (SHAP), and Local Interpretable Model-agnostic Explanations (LIME) to analyse model behaviour in chest X-ray pneumonia classification using ResNet-50.

While the model achieved strong classification performance (AUC = 0.9701, PR-AUC = 0.9796), cross-method analysis revealed that explanation consistency varies significantly across cases — particularly in failure cases. Grad-CAM exhibited gradient saturation on high-confidence predictions, SHAP revealed decision boundary conflicts in false negative cases, and LIME identified coarse segment-level contributions with high variance.

We demonstrate that disagreement between XAI methods can reveal unreliable model reasoning, even when predictions are correct. These findings suggest that cross-method agreement may serve as a practical and interpretable indicator of trust in medical AI systems — a framework not previously formalised in the chest X-ray classification literature. In high-stakes medical AI, agreement between explanations may matter more than accuracy alone.

1. Introduction

The deployment of deep learning models in healthcare has grown rapidly over the past decade, with convolutional neural networks demonstrating performance comparable to clinical experts on tasks such as chest X-ray interpretation. However, the black-box nature of these models presents a fundamental barrier to clinical adoption. In safety-critical settings, a prediction without an explanation is insufficient. Clinicians require not only accurate diagnoses, but also transparent reasoning that can be audited, challenged, and trusted.

Explainability methods have been proposed to address this limitation. Gradient-weighted Class Activation Mapping (Grad-CAM) produces spatial heatmaps highlighting the regions most influential to a prediction. SHapley Additive exPlanations (SHAP) assigns contribution scores to regions of the image, quantifying the direction and magnitude of each region's influence. Local Interpretable Model-agnostic Explanations (LIME) approximates model behaviour locally by perturbing image segments and fitting a linear model to the results. Each of these methods has been applied individually in medical imaging contexts, demonstrating the potential of XAI for clinical transparency.

However, a critical gap remains. Existing work typically applies a single XAI method in isolation, without evaluating whether that method faithfully represents the model's decision process or whether its explanations are consistent with those produced by alternative methods. To the best of our knowledge, no prior work has systematically compared Grad-CAM, SHAP, and LIME on identical predictions to evaluate cross-method agreement as a measure of explanation reliability in chest X-ray classification.

This study addresses this gap by proposing a multi-method explainability pipeline for pneumonia detection using ResNet-50. The main contributions of this study are as follows. First, we implement and compare three complementary XAI methods on identical chest X-ray images. Second, we introduce cross-method agreement as a framework for evaluating trust in model explanations — demonstrating that consistent explanations across methods are associated with more reliable reasoning, while disagreement reveals potential failure cases. Third, we conduct a detailed false negative analysis using all three methods simultaneously, providing mechanistic explanations for cases where the model failed to detect pneumonia. This work reframes explainability in medical AI as a multi-perspective validation problem rather than a single-method visualisation task.

The remainder of this paper is structured as follows. Section 2 reviews related literature on medical image classification and explainable AI. Section 3 describes the methodology including dataset, model architecture, training setup, and XAI implementation. Section 4 presents quantitative and qualitative results. Section 5 discusses key findings and limitations. Section 6 concludes with directions for future work.

2. Related Work

2.1 Medical Image Classification using Deep Learning

The use of convolutional neural networks for medical image classification has grown significantly over the past decade. He et al. (2016) introduced **ResNet**, demonstrating that deep residual networks with skip connections could achieve state-of-the-art performance on image classification tasks. This architecture has since become the foundation for medical imaging applications due to its ability to train very deep networks without degradation.

Rajpurkar et al. (2017) demonstrated that a deep learning model trained on chest X-rays could detect pneumonia at a level comparable to practising radiologists, achieving an AUC of 0.769. This work established the clinical potential of CNN-based diagnosis but did not address model transparency or Explainability.

Irvin et al. (2019) released CheXpert, a large chest X-ray dataset with uncertainty labels, and demonstrated that models trained on such datasets could generalise across clinical settings. However, the black-box nature of these models remained a concern for clinical adoption. The common limitation across these works is that high classification accuracy does not imply clinical trustworthiness — models may rely on spurious correlations, and without explainability tools this cannot be detected.

2.2 Explainable AI in Healthcare

Selvaraju et al. (2017) introduced Gradient-weighted Class Activation Mapping **(Grad-CAM)**, a technique that uses gradients flowing into the final convolutional layer to produce a coarse localisation heatmap highlighting the regions most influential to the prediction. Grad-CAM has been widely adopted in medical imaging due to its simplicity and visual interpretability. However, it produces low-resolution maps and can suffer from gradient saturation in high-confidence predictions — a limitation directly observed in this project.

Lundberg and Lee (2017) introduced **SHAP** (SHapley Additive exPlanations), grounded in cooperative game theory. SHAP assigns a contribution score to each feature — in image classification, each pixel — quantifying how much it pushed the prediction toward or away from a given class. Unlike Grad-CAM, SHAP provides directional information (positive and negative contributions) making it more informative for understanding decision conflicts.

Ribeiro et al. (2016) introduced **LIME** (Local Interpretable Model-agnostic Explanations), which approximates model behaviour locally by perturbing input features and fitting a linear model to the results. LIME is model-agnostic and produces human-friendly segment-level explanations, but exhibits high variance due to its perturbation-based sampling — running it multiple times on the same image can produce different results.

A key finding across these works is that each method captures a different aspect of model behaviour. No single method provides a complete explanation, which motivates the multi-method approach adopted in this project.

2.3 Comparison of XAI Methods

Despite the widespread adoption of individual XAI methods, comparative studies remain limited. Samek et al. (2017) evaluated several explanation methods by measuring how much model performance dropped when the most important pixels were removed — a metric known as the pixel flipping test. They found that different methods produced significantly different importance maps on the same input, raising questions about which method best reflects true model reasoning.

Arun et al. (2021) compared **Grad-CAM**, Integrated Gradients, and **SHAP** on chest X-ray classification tasks and found that no single method consistently aligned with radiologist annotations. This suggests that explanation faithfulness — whether an explanation truly reflects the model's decision process — cannot be assumed without validation.

Ghassemi et al. (2021) argued that explainability methods in clinical AI are often evaluated in isolation and that disagreement between methods is a strong signal of unreliable model reasoning. They called for multi-method validation frameworks before clinical deployment of AI systems. The gap identified across these works is clear: most studies either use a single XAI method or compare methods visually without a systematic trust evaluation framework. This project directly addresses that gap by analysing agreement and disagreement between **Grad-**

CAM, **SHAP**, and **LIME** as a proxy for model trustworthiness — a direction that remains underexplored in existing literature.

2.4 Research Gap & Contribution

The reviewed literature highlights three key limitations. First, most medical imaging studies prioritise classification performance without addressing model transparency. Second, explainability research typically relies on a single XAI method, without evaluating whether the explanations faithfully represent model reasoning. Third, comparative studies of XAI methods rarely provide a systematic framework for assessing trust in model decisions.

To the best of our knowledge, limited work has explored cross-method agreement between **Grad-CAM**, **SHAP**, and **LIME** as a mechanism for evaluating trust in chest X-ray classification.

This project addresses these limitations by proposing a multi-method explainability framework based on agreement and disagreement between XAI techniques. We demonstrate that consistent explanations across methods are associated with more reliable model reasoning, while disagreement can reveal potential failure cases.

Our central hypothesis is:

"Agreement between XAI methods can serve as a proxy for trust in medical AI systems."

3. Methodology

3.1 Dataset

The Chest X-Ray Images (Pneumonia) dataset from Kaggle was used, containing 5,216 training images, 624 test images, and 16 validation images across two classes — Normal and Pneumonia. The dataset is imbalanced, with approximately 73% pneumonia cases in the training set. This imbalance was addressed through weighted sampling and augmentation during training.

3.2 Preprocessing

Images were resized to 224×224 pixels to match ResNet-50 input requirements. Training augmentations included random horizontal flips, rotation up to 10 degrees, and colour jitter. All images were normalised using dataset-specific statistics — mean 0.5330 and standard deviation 0.0349 — computed from the training set rather than standard ImageNet values, to better represent the greyscale X-ray distribution.

3.3 Model Architecture

A ResNet-50 architecture was trained from scratch for binary classification. The standard fully connected head was replaced with a custom head consisting of a dropout layer (p=0.5), a linear layer reducing to 256 dimensions, ReLU activation, a second dropout layer (p=0.3), and a final linear layer with a single sigmoid output. This architecture was selected for its strong performance on medical imaging tasks and its well-studied layer structure, which is compatible with gradient-based explainability methods.

3.4 Training Setup

The model was trained for 10 epochs using Binary Cross-Entropy loss, Adam optimiser with a learning rate of 1e-4, and a StepLR scheduler reducing the learning rate by a factor of 0.1 every 3 epochs. The best checkpoint was selected based on validation AUC. Final test performance achieved AUC = 0.9701 and PR-AUC = 0.9796.

3.5 XAI Methods

Three complementary explainability methods were applied to the trained model:

Grad-CAM (Selvaraju et al., 2017) — Forward and backward hooks were registered on layer4[2].conv3 of ResNet-50. Gradients were globally average-pooled to produce per-channel importance weights, and a weighted sum of activation maps passed through ReLU produced the localisation heatmap.

SHAP (Lundberg and Lee, 2017) — PartitionExplainer with a blur masker baseline was used. SHAP values quantify the direction and magnitude of each pixel's contribution to the final prediction. Positive values indicate features pushing toward pneumonia, negative values indicate features pushing toward normal.

LIME (Ribeiro et al., 2016) — LimeImageExplainer was applied with 1000 perturbation samples and 10 superpixel features. LIME fits a local linear model to identify which image segments most influenced the prediction.

All explanations were generated on the test set to ensure evaluation on unseen data.

3.6 Trust Evaluation Framework

A key methodological contribution of this project is the use of cross-method agreement as a proxy for trust. When Grad-CAM, SHAP, and LIME highlight consistent regions, confidence in the explanation — and potentially the model's reasoning — is higher. When methods disagree, the model's decision is flagged as potentially unreliable. This framework was applied to both correctly classified cases and false negatives to evaluate explanation consistency across different prediction outcomes.

This framework enables qualitative evaluation of model reasoning beyond traditional performance metrics.

4. Results

4.1 Quantitative Results

The trained ResNet-50 model achieved strong classification performance on the held-out test set of 624 images. The key metrics are as follows:

- AUC (Area Under the ROC Curve): 0.9701
- PR-AUC (Area Under the Precision-Recall Curve): 0.9796
- Optimal classification threshold: 0.5 (selected based on F1 score maximisation during threshold analysis)

The high PR-AUC of 0.9796 is particularly significant given the class imbalance in the dataset — approximately 73% pneumonia cases in training. PR-AUC is a more informative metric than AUC in imbalanced settings as it directly measures the trade-off between precision and recall for the minority class.

At the default threshold of 0.5, the model correctly identified the majority of pneumonia cases while maintaining low false positive rates. However, threshold analysis revealed 3 false negatives out of 234 pneumonia test cases (recall = 0.987) — pneumonia cases the model failed to detect — which became the primary focus of the explainability analysis in subsequent sections.

The model demonstrates strong performance, approaching levels reported in prior studies, but further validation on external datasets is required before clinical applicability can be claimed. Classification accuracy alone is insufficient for clinical deployment — the following sections evaluate whether the model's reasoning process can be trusted.

4.2 Qualitative Results — XAI Findings

4.2.1 Grad-CAM

Grad-CAM analysis revealed that the model's spatial attention varied significantly across cases. For correctly classified pneumonia cases, Grad-CAM highlighted the lower and central lung fields — consistent with the anatomical regions affected by bacterial consolidation. For normal cases, activation was diffuse and low, indicating no dominant focus region.

However, two important limitations were observed. First, on high-confidence pneumonia predictions ($p = 1.000$), Grad-CAM produced near-zero variance activation maps — a phenomenon known as gradient saturation. This means Grad-CAM failed to produce a meaningful explanation precisely when the model was most certain. Second, in misclassified normal cases, Grad-CAM highlighted the lower lung and abdominal region — not a clinically

relevant area for pneumonia diagnosis — suggesting spatial bias in the model's attention mechanism.

(Figure 1 — Grad-CAM heatmap overlay)

Figure 1 shows activation concentrated in the lower lung region for pneumonia cases, with near-zero activation for normal cases.

4.2.2 SHAP

SHAP analysis provided directional attribution maps showing which regions pushed the prediction toward or away from pneumonia. For correctly classified pneumonia cases, SHAP identified strong positive contributions in the central and lower lung fields — consistent with consolidation patterns in bacterial pneumonia. For normal cases, SHAP assigned predominantly negative values across the image, correctly suppressing the pneumonia prediction.

In false negative cases, SHAP revealed a decision boundary conflict — the model detected some pneumonia signal in the upper lung fields but was simultaneously suppressed by strong negative contributions from other regions, resulting in a net prediction below the 0.5 threshold. This mechanistically explains why these cases were missed.

(Figure 2 — SHAP contribution map)

Figure 2 shows positive red contributions in the central lung fields for pneumonia and predominantly negative blue values for normal cases.

4.2.3 LIME

LIME identified coarse superpixel regions contributing to each prediction. For pneumonia cases, LIME highlighted the right lung field as the dominant contributing segment. For normal cases, LIME suppressed lung field segments and highlighted peripheral regions. However, LIME exhibited high variance due to its perturbation-based sampling — a known limitation of segment-based explanation methods.

(Figure 3 — LIME segment explanation)
Figure 3 shows highlighted superpixel segments in the right lung field for pneumonia and peripheral suppression for normal cases.

4.2.4 False Negative Analysis

The three false negative cases were analysed using all three XAI methods simultaneously. A consistent pattern emerged across all methods: the model failed when pneumonia was distributed across multiple lung regions rather than concentrated in a single high-contrast area. This suggests the model has developed a bias toward localised features — a known risk when training on datasets dominated by bacterial pneumonia cases with dense focal consolidations.

4.3 Comparative Results — Cross-Method Agreement Analysis

The three XAI methods were applied to identical images to evaluate cross-method agreement — the core research contribution of this project.

(Figure 4 — three-way XAI comparison grid)

Figure 4 presents all three methods side by side on identical images, enabling direct visual comparison of spatial agreement and disagreement.

Table 1: Cross-Method XAI Agreement Summary

Case	Prediction	Grad-CAM	SHAP	LIME	Agreement
Normal	P = 0.105	Low/diffuse	Negative	Suppressive	Strong Agreement
Pneumonia	P = 1.000	Flat (saturated)	Positive	Positive	Partial Agreement
FN Case 1	P = 0.043	Peripheral	Negative	Peripheral	Strong Agreement
FN Case 2	P = 0.215	Lower left	Suppressed	Peripheral	Partial Agreement
FN Case 3	P = 0.098	Lateral wall	Negative	Peripheral	Strong Agreement

4.3.1 Agreement Cases

For the normal image (p = 0.105), all three methods were consistent:
- Grad-CAM showed diffuse low activation across lung fields
- SHAP assigned predominantly negative values across the image
- LIME highlighted peripheral and shoulder regions as suppressive

This three-way agreement provides higher confidence that the model's reasoning is reliable for this case. When all methods point to the same conclusion, the explanation can be considered more faithful to the model's actual decision process.

4.3.2 Disagreement Cases

For the pneumonia image (p = 1.000), a critical conflict emerged between methods:
- SHAP showed strong positive contributions in the central and lower lung fields — consistent with clinical anatomy
- LIME identified lung field segments as positively contributing
- Grad-CAM produced a completely flat activation map due to gradient saturation

To quantify this disagreement, Pearson correlation was computed between the Grad-CAM and SHAP maps. For the normal image, r = 0.4236, indicating moderate agreement. For the pneumonia image, Grad-CAM produced a near-zero variance map, making correlation mathematically undefined and confirming complete attribution collapse.

(Figure 5 — full XAI pipeline demo)

Figure 5 shows the complete pipeline output — prediction, Grad-CAM, SHAP, and LIME — for a single pneumonia case.

4.3.3 Trust Evaluation

These findings support the central hypothesis of this project:

"Agreement between XAI methods can serve as a proxy for trust in medical AI systems."

When all three methods agree → higher confidence in explanation reliability.
When methods disagree → the model's decision should be treated with caution.

Critically, disagreement between XAI methods does not always indicate a wrong prediction — the pneumonia case was correctly classified at p = 1.000 despite Grad-CAM's failure. The model was correct. The explanation was not. This mismatch raises serious concerns for clinical deployment of single-method interpretability pipelines.

5. Discussion

5.1 Model Performance & Clinical Relevance

The model achieved AUC = 0.9701 and PR-AUC = 0.9796, demonstrating clinically competitive performance on the chest X-ray pneumonia detection task. However, the presence of 3 false negatives — cases where pneumonia was missed entirely — highlights that high aggregate performance does not guarantee reliability at the individual case level. In clinical settings, a single missed pneumonia diagnosis can have serious consequences, making false negative analysis critical.

5.2 Spatial Bias in Model Attention

Grad-CAM analysis revealed that the model occasionally attends to non-anatomical regions — including the lower lung border and abdominal area — when misclassifying normal cases as pneumonia. This spatial bias suggests the model may have learned dataset-specific artifacts rather than purely pathological features. This is consistent with findings in the literature — Arun et al. (2021) demonstrated that explanation maps frequently fail to align with radiologist annotations, even when classification accuracy is high.

5.3 Decision Boundary Conflict

SHAP analysis of false negative cases revealed a decision boundary conflict — the model detected pneumonia signal in some lung regions but was simultaneously suppressed by negative contributions from other regions. This conflict, confirmed independently by LIME, provides a mechanistic explanation for why these cases were missed. It suggests the model is sensitive to the spatial distribution of disease — performing well on focal consolidations but struggling with diffuse or bilateral patterns.

5.4 XAI Method Reliability

The comparative analysis demonstrated that different XAI methods capture fundamentally different aspects of model behaviour:

- Grad-CAM systematically fails to capture diffuse pathological patterns and suffers from gradient saturation on high-confidence predictions. It should not be used as the sole explanation method in clinical deployment.

- SHAP demonstrated the most consistent behaviour in this study, suggesting it may be more suitable for clinical audit scenarios compared to gradient-based methods.

- LIME provides intuitive segment-level explanations but exhibits high variance due to perturbation-based sampling. It is useful for demonstration but less reliable for rigorous evaluation.

5.5 Explanation Faithfulness & Trust

A critical finding of this project is that model correctness does not guarantee explanation reliability. The pneumonia case was correctly classified at $p = 1.000$, yet Grad-CAM produced a completely flat activation map — providing no useful explanation despite the correct prediction. This highlights the issue of explanation faithfulness and raises concerns about relying on a single XAI method in safety-critical applications.

To our knowledge, this is one of the first works to explicitly evaluate cross-method agreement between Grad-CAM, SHAP, and LIME as a proxy for trust in medical imaging models.

These findings reinforce the importance of evaluating explanation consistency across multiple methods. This project proposes cross-method agreement as a validation step before clinical trust is established. This framework has not been formally explored in prior chest X-ray classification studies.

5.6 Limitations

Several limitations of this work should be acknowledged:

- The XAI comparison was conducted on a limited number of test images. Statistical generalisation requires evaluation across a larger dataset.
- LIME's instability means results may vary across runs, limiting reproducibility.
- The dataset is dominated by bacterial pneumonia cases, which may bias the model toward localised consolidation patterns and reduce performance on viral or atypical pneumonia presentations.
- The trust evaluation framework proposed here is qualitative. Future work should formalise this framework by defining quantitative agreement metrics — such as correlation scores and spatial overlap measures — and thresholds for trust calibration.

6. Conclusion

This project presented a multi-method explainability pipeline for chest X-ray pneumonia classification using ResNet-50. The pipeline combines Grad-CAM, SHAP, and LIME to provide complementary explanations of model decisions, and introduces cross-method agreement as a framework for evaluating trust in medical AI systems.

The model achieved strong classification performance with AUC = 0.9701 and PR-AUC = 0.9796. However, the primary contribution of this work is not the classification performance — it is the systematic analysis of when and why the model's explanations can be trusted.

Three key findings emerged from this analysis:

First, Grad-CAM systematically fails to capture diffuse pathological patterns and suffers from gradient saturation on high-confidence predictions — making it unreliable as a standalone explanation method in clinical settings.

Second, SHAP demonstrated the most consistent behaviour in this study, suggesting it may be more suitable for clinical audit scenarios compared to gradient-based methods.

Third, and most importantly — disagreement between XAI methods is a strong indicator of unreliable model reasoning. A model can be correct for the wrong reasons, and only a multi-method XAI approach can expose this.

This finding directly supports the central hypothesis of this project:

"Agreement between XAI methods can serve as a proxy for trust in medical AI systems."

This work demonstrates that explainability in medical AI should not be treated as a single-visualisation problem, but as a multi-perspective validation process. Future work should evaluate cross-method agreement quantitatively across larger datasets, incorporate radiologist feedback to validate explanation faithfulness, and extend the framework to other medical imaging modalities and architectures. In high-stakes medical AI, agreement between explanations may be as critical as accuracy itself.

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The code, notebooks, and full experimental pipeline are publicly available at:
https://github.com/emanaak04-svg/medical-xai